CHRISTIAN SOCIAL SERVICES COMMISSION (CSSC)
NORTHERN ZONE JOINT EXAMINATIONS SYNDICATE (NZ-JES)

FORM FOUR PRE – NATIONAL EXAMINATION

032/2A

CHEMISTRY 2A
(ACTUAL PRACTICAL A)

Time: 2:30 Hours **Thursday 27th August, 2026 a.m.**

Instructions

1. This paper consists of **two (2)** questions. Answer **all** questions.
2. Each question carries **twenty five (25) marks.**
3. Qualitative Analysis Guide sheets of NECTA and non programmable calculators may be used.
4. Cellular phones, smart watch and any other unauthorized materials are not allowed in the Examination room.
5. Write your **Examination number** on every page of your answer booklet(s) provided.
6. You may use the following constants:

Atomic masses:

H=1, C= 12, O = 16, Na= 23, N=14, Ca=40 Cl=35.5 S=32 .

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1. 20 litres of waste water from laboratory drainage system was collected using the trash bin. This waste water was contaminated with nitric acid (HNO_3) making the solution acidic. But before disposing this waste solution it was supposed to be neutralized by marble chips. But you were told to first determine the concentration of the waste acidic solution using sodium hydroxide solution.

You are provided with the following solutions;

- **Solution A**: Containing waste water contaminated with nitric acid.
- **Solution B**: Containing 3.0g of sodium hydroxide per 1.5L of solution.
- Phenolphthalein indicator (**POP**)

Procedure:

(i) Pour the solution **A** into the burette.

(ii) Pipette 20 ml or 25ml of solution **B** and put into a titration flask then add three drops of phenolphthalein indicator.

(iii) Titrate this solution **A** against the solution **B** until the end point reaches. Repeat the titration to obtain three more values and record your results in tabular form.

(a) Complete the table by filling the titre values:-

(i) Calculate the average volume of acid used.

(ii) Write down a balanced chemical equation for the reaction between **A** and **B**.

(b) Calculate the molarity of:

(i) Solution **B**.

(ii) Solution **A**.

(c) (i) Determine the number of moles present in 20 litres of waste acid.

(ii) How many grams of marble chips are required to neutralize the waste acidic solution before disposing it?

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2. You are provided with the following:-

Solution **Z** containing 1M sodium thiosulphate ($Na_2S_2O_3$);

Solution **T** containing 0.1M nitric acid (HNO_3);

Distilled water.

Piece of paper marked **X**;

Stop-watch.

Procedure:-

(i) Using measuring cylinder, measure 5cm³ of solution **Z** and put into 100 cm³ beaker.

(ii) Add distilled water into the beaker containing solution **Z** as shown in **Table 1.**

(iii) Measure 5cm³ of solution **T** and put into 100 cm³ beaker containing solution **Z**, and immediately start the stop-watch.

(iv) Swirl the contents in the 100cm³ beaker and put the beaker on top of mark **X** on the piece of paper. Watch the mixture from above and observe changes.

(v) Switch off the stop-watch when the mark **X** disappears.

(vi) Record the time taken for the letter **X** to disappear.

(vii) Repeat procedures (i) to (vi) using the data shown in **Table 1**.

Table 1.

Experiment	Vol. of T (cm^3)	Vol. of Z (cm^3)	Vol. of Distilled Water (cm^3)	Time (s)	Rate (s^{-1})
1	5	5	0		
2	5	4	1		
3	5	3	2		
4	5	2	3		

Questions:

(a) Complete **Table 1**.

(b) Write a balanced equation for reaction between **T** and **Z**.

(c) What substance was produced during the reaction which obscured letter **X**?

(d) Plot the graph of volume of $Na_2S_2O_3$ solution against time(s).

(e) What conclusion can you draw from this experiment?

(f) Explain **two (2)** factors that affect the rate of chemical reaction.

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